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216 B. Yılmaz

1a) $\frac{4}{3}\pi R^3$ $\quad Q$

$\frac{4}{3}\pi\left(\frac{R}{2}\right)^3$ $\quad Q_{iç}$

$$\Rightarrow \cancel{\frac{4}{3}}\cancel{\pi}\cancel{R^3}\, Q_{iç} = \cancel{\frac{4}{3}}\cancel{\pi}\cdot\frac{1}{8}\cdot\cancel{R^3}\cdot Q$$

$$\Rightarrow \boxed{Q_{iç} = \frac{Q}{8}}$$

1b) $r = \frac{R}{2}$

$$\oint E\,da = \frac{Q_{iç}}{\varepsilon_0}$$

$$\oint E\,(4\pi r^2) = \frac{Q}{8\varepsilon_0}$$

$$E = \frac{4}{32\pi\varepsilon_0 r^2} = \frac{Q}{R^2}\cdot\frac{1}{8\pi\varepsilon_0}$$

$r = R$

$$\oint E\,(4\pi r^2) = \frac{Q}{\varepsilon_0}$$

$$E = \frac{Q}{R^2}\cdot\frac{1}{4\pi\varepsilon_0}$$

$$\frac{E_{\frac{R}{2}}}{E_R} = \frac{\frac{\cancel{Q}}{\cancel{R^2}}\cdot\frac{1}{2\cancel{8\pi\varepsilon_0}}}{\frac{\cancel{Q}}{\cancel{R^2}}\cdot\frac{1}{\cancel{4\pi\varepsilon_0}}} = \boxed{\frac{1}{2}}$$

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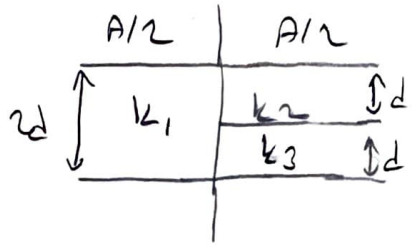
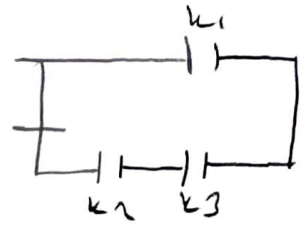
2)





$A = 16,5\ cm^2$
$2d = 7,12\ mm$
$k_1 = 21$
$k_2 = 42$
$k_3 = 58$

$C_{es} = 8,85 \cdot 10^{-12}\ C^2/Nm^2$

$$C = \frac{A \cdot \varepsilon_0}{d}$$

$$\frac{1}{C_{23}} = \frac{1}{C_2} + \frac{1}{C_3} \Rightarrow C_{23} = \frac{C_2 \cdot C_3}{C_2 + C_3}$$

$$C = C_1 + \frac{C_2 C_3}{C_2 + C_3}$$

$$= \frac{k_1 \cdot A/2 \cdot \varepsilon_0}{2d} + \frac{\left(\frac{k_2 A/2\, \varepsilon_0}{d}\right)\left(\frac{k_3 A/2\, \varepsilon_0}{d}\right)}{\frac{k_2 A/2\, \varepsilon_0 + k_3 A/2\, \varepsilon_0}{d}}$$

$$\frac{21 \cdot 10,5 \cdot 10^{-4} \cdot 8,85 \cdot 10^{-12}}{4 + \frac{7,12 \cdot 10^{-3}}{2}} + \frac{42 \cdot 58 \cdot 10,5 \cdot 10^{-4} \cdot 8,85 \cdot 10^{-12}}{7,12 \cdot 10^{-3} \cdot 100}$$

$$C = 1,37 \cdot 10^{-11} + 3,178 \cdot 10^{-11} \Rightarrow \boxed{C = 4,55 \cdot 10^{-11}\ F}$$

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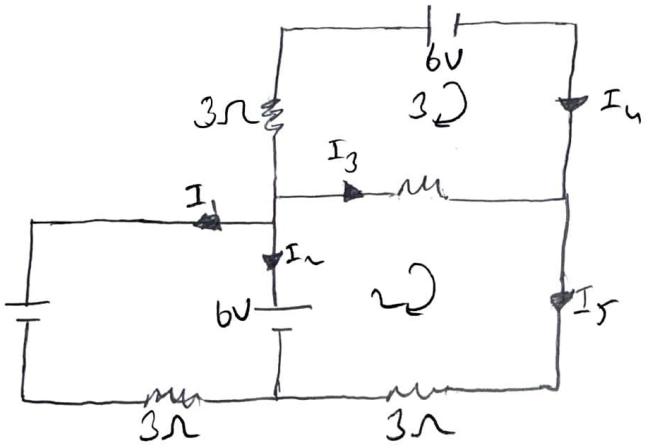
3)



$I_3 + I_4 = I_5$ (1)

$I_1 + I_2 = I_3 + I_4$ (2)

(1) $6 - 6 = -3I_1$
$I_1 = 0$ //

(2) $-6 = -3I_3 + 3I_4$
$2 = I_3 - I_4$

(3) $6 = 3I_3 + 3I_5$
$I_5 = 2 - I_3$

(1)

$I_3 + I_4 = I_5$

$I_3 + (I_3 - 2) = 2 - I_3$

$3I_3 = 4 \Rightarrow I_3 = \frac{4}{3}$ A //

$I_4 = I_3 - 2 = \frac{4}{3} - 2 = -\frac{2}{3}$ A (Ters yön)

$I_5 = 2 - I_3 = 2 - \frac{4}{3} = \frac{2}{3}$ A //

$I_1 + I_2 = I_3 + I_4$

$0 + I_2 = \frac{4}{3} - \frac{2}{3} \Rightarrow I_2 = \frac{2}{3}$ A //

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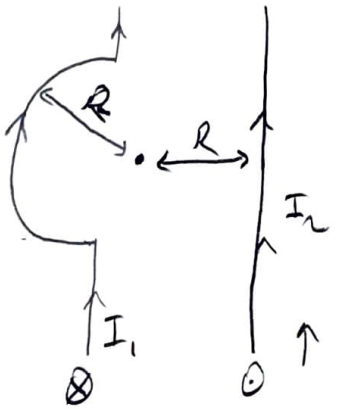
4)



$\vec{B}_1$ manyetik alan ⊗ ise doğru
$\vec{B}_2$ manyetik alan ⊙ dışa doğru

Akım yönü ↑ olmalı.

$$\vec{B}_1 = K\,\frac{2i\pi}{R}\cdot\frac{1}{2} \qquad \vec{B}_2 = K\,\frac{2i}{R}$$

Manyetik alan 0 olabilmesi için nötrlemelidir.

$$\vec{B}_1 = \vec{B}_2 \Rightarrow \frac{1}{2}\cdot K\cdot\frac{2i_1\pi}{R} = K\cdot\frac{2i_2}{R}$$

$$i_1\frac{\pi}{2} = i_2 \Rightarrow \boxed{i_2 = \frac{\pi}{2}\,A}$$

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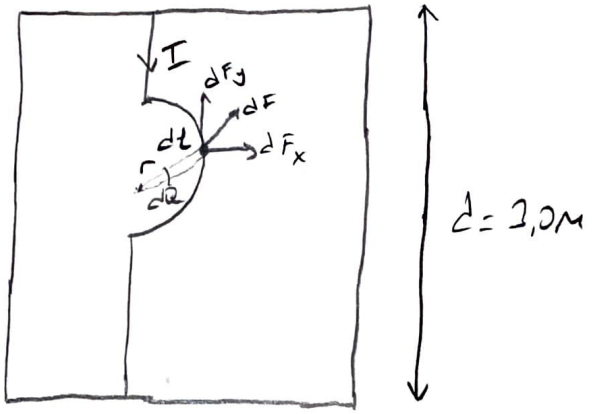
5)



$r = 0,95M$
$I = 3,40A$
$B = 2,20T$

Yarım daire üzerine etki eden manyetik kuvvet (F_1)

$dl = R.dQ$
$F = I.dl.B_0$
$= I.R.dQ.B_0$

dl ile B_0 arasındaki açı 90°'dir.

$\int dF_y = \int dF.\sin Q$

$F_y = I.R.B_0.\int dQ \sin Q$
$F_y = I.R.B_0.(-\cos Q)\Big|_0^{\pi}$
$\boxed{F_y = 2I.B_0.R}$

$\int dF_x = \int dF.\cos Q$

$F_x = I.B_0.2.\int dQ \cos Q$
$F_x = I.B_0.R.\sin Q\Big|_0^{\pi}$
$\boxed{F_x = 0}$

$F_1 = \sqrt{F_x^2 + F_y^2} = 2IB_0R = 2.3,40.2,20.0,95$
$\boxed{F_1 = 14,212\ N}$

Telin geri kalanına etki eden manyetik kuvvet (F_2)

Tel uzunluğu (l) $= d - 2r$

$F_2 = I.l.B_0$
$= I.(d-2r).B_0 = 3,40.(3-2.0,95).2,20$
$\boxed{F_2 = 8,228\ N}$

$F_{Net} = F_1 + F_2 = \boxed{22,44\ N}$